

EVM-Private "Code Updates"

Rome

HALBORN

Summary

100% OF ALL REPORTED FINDINGS HAVE BEEN ADDRESSED

ABOUT THIS REPORT

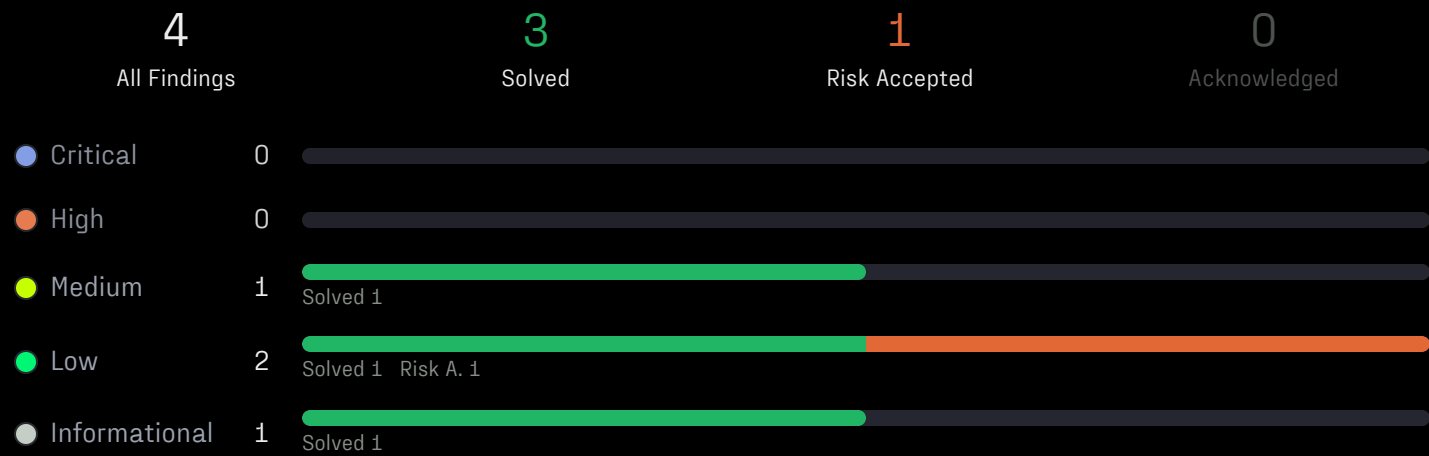
ASSESSMENT TYPE
Assurance Assessment

SOURCE CODE
🔗 rome-evm-private

DATE OF ENGAGEMENT
Jul 29, 2025 - Aug 16, 2025

ASSESSED COMMIT ID
🔗 1acc83f

SEVERITY BREAKDOWN



Introduction

Rome engaged Halborn to conduct a security assessment on their Rome EVM program beginning on July 29, 2025 and ending on August 15, 2025. The security assessment was scoped to the smart contracts provided in the GitHub repository rome-evm-private, commit hashes, and further details can be found in the Scope section of this report.

The Rome team is releasing a new version their Rome EVM program Solana program. This program collectively allows for functioning as an EVM-compatible execution layer on the Solana blockchain, enabling the deployment and execution of Solidity smart contracts. This new version includes new features, such as the use of Address Lookup Tables and the possibility of depositing SPL Tokens.

Assessment Summary

Halborn was provided 2 weeks for the engagement and assigned one full-time security engineer to review the security of the Solana Programs in scope. The engineer is a blockchain and smart contract

security expert with advanced smart contract hacking skills, and deep knowledge of multiple blockchain protocols.

The purpose of the assessment is to:

- Identify potential security issues within the Solana Program.
- Ensure that smart contract functionality operates as intended.

In summary, Halborn identified some improvements to reduce the likelihood and impact of risks, which were mostly addressed by the **Rome team**. The main ones were the following:

- Implement the `spl_token` program as part of the emulation for Withdraw operation.
- Avoid to use the deprecated `SlotHashesSysvar` in favor of `PodSlotHashes` to check if an alt can be closed.
- Increment the computation unit of a transaction when calling the `Withdraw` function and the destination token account is not created.

Scope

REPOSITORY

(a) Repository:  **rome-evm-private**

(b) Assessed Commit ID: `1acc83f496d3075bb300465c72032dad6837192e`

(c) Items in scope:

- ✓  `program/src` 80 files
- ✓  `accounts` 17 files
 - <> `account_state.rs` Rust
 - <> `account_type.rs` Rust
 - <> `alt_id.rs` Rust
 - <> `alt_slots.rs` Rust
 - <> `code.rs` Rust
 - <> `holder.rs` Rust
 - <> `lock.rs` Rust
 - <> `mod.rs` Rust
 - <> `owner_info_v0.rs` Rust
 - <> `owner_info.rs` Rust
 - <> `ro_lock.rs` Rust
 - <> `slot.rs` Rust
 - <> `state_holder.rs` Rust
 - <> `storage.rs` Rust
 - <> `tx_holder.rs` Rust
 - <> `valids.rs` Rust

- <> ver.rs Rust
- ✓  api 10 files
 - <> alt_alloc.rs Rust
 - <> alt_dealloc.rs Rust
 - <> deposit.rs Rust
 - <> do_tx_holder_iterative.rs Rust
 - <> do_tx_holder.rs Rust
 - <> do_tx_iterative.rs Rust
 - <> do_tx.rs Rust
 - <> mod.rs Rust
 - <> reg_owner.rs Rust
 - <> transmit_tx.rs Rust
- ✓  context 4 files
 - <> atomic.rs Rust
 - <> iterative_lock.rs Rust
 - <> iterative.rs Rust
 - <> mod.rs Rust
- ✓  non_evm 10 files
 - <> aspl_token_ix.rs Rust
 - <> aspl_token.rs Rust
 - <> aux.rs Rust
 - <> mod.rs Rust
 - <> non_evm_state.rs Rust
 - <> spl_token_ix.rs Rust
 - <> spl_token.rs Rust
 - <> system_ix.rs Rust
 - <> system.rs Rust
 - <> withdraw.rs Rust
- ✓  precompile 10 files
 - <> blake2f.rs Rust
 - <> ecadd.rs Rust
 - <> ecmul.rs Rust
 - <> ecpairing.rs Rust
 - <> ecrecover.rs Rust
 - <> identity.rs Rust
 - <> mod.rs Rust
 - <> modexp.rs Rust
 - <> ripemd_160.rs Rust
 - <> sha2_256.rs Rust
- ✓  state 11 files
 - <> allocate.rs Rust
 - <> aux.rs Rust
 - <> base.rs Rust
 - <> handler.rs Rust
 - <> info.rs Rust
 - <> journal.rs Rust
 - <> journaled_state.rs Rust
 - <> mod.rs Rust

```
<> origin.rs Rust
<> pda.rs Rust
<> state.rs Rust
v tx 6 files
  <> deposit.rs Rust
  <> eip1559.rs Rust
  <> eip2930.rs Rust
  <> legacy.rs Rust
  <> mod.rs Rust
  <> tx.rs Rust
v vm 5 files
  <> mod.rs Rust
  <> snapshot.rs Rust
  <> vm_atomic.rs Rust
  <> vm_iterative.rs Rust
  <> vm.rs Rust
<> alloc.rs Rust
<> assert.rs Rust
<> config.rs Rust
<> entrypoint.rs Rust
<> error.rs Rust
<> lib.rs Rust
<> migrate.rs Rust
```

Out-of-Scope: Third party dependencies and economic attacks.

REMEDATION COMMIT ID:

```
5c2712c578fca198da617c13d33bf938752ff582
a45a4844df27e5c4f6e2bf63cf812192a71ba101
11561c3b4ccb71278836a7631d2ec7bee156aa6d
```

Out-of-Scope: New features/implementations after the remediation commit IDs.

Findings Overview

#	TITLE	SEVERITY	SCORE	STATUS
HAL-001	Incomplete Precompile Emulation Causes Panic on SPL Token Withdrawal	● Medium	5.6	SOLVED 08/22/2025

#	TITLE	SEVERITY	SCORE	STATUS
HAL-002	Repeated Sysvar Deserialization in Loop Leads to Heap Exhaustion	● Low	2.8	SOLVED 09/03/2025
HAL-003	Insufficient Compute Unit Budget Causes Withdrawal Failures	● Low	2.0	RISK ACCEPTED 09/02/2025
HAL-004	Attempting to Extend a Deactivated Address Lookup Table Causes Allocation to Fail	● Informational	1.9	SOLVED 09/03/2025

Disclaimer

Halborn strongly recommends conducting a follow-up assessment of the project either within six months or immediately following any material changes to the codebase, whichever comes first. This approach is crucial for maintaining the project's integrity and addressing potential vulnerabilities introduced by code modifications.

